

Diagnostic information - Headlight control unit

Stepper motor for dyn. cornering light, position sensor

CAUTION

Danger of injury from high voltage!

- ◆ The light must always be switched off when removing and installing the headlights.
- ◆ The gas-discharge lamp may only be operated when installed in the headlight.
- ◆ Only carry out repairs to the headlight or wiring when the part is removed and disconnected from the power supply.

INFORMATION

The stepper motor/servo motor for the dynamic cornering light and its mechanical components and wiring cannot be replaced individually.
The headlight housing must be replaced if repairs are needed.

Diagnostic conditions

- Internal function test on headlight (Xenon) control unit
- Ignition on

Possible fault causes

- ◆ Mechanism of stepper motor for dyn. cornering light sluggish or blocked
- ◆ Dynamic cornering light wiring faulty
- ◆ Stepper motor for dynamic cornering light faulty
- ◆ Headlight control unit faulty

Diagnostic information - Headlight control unit

Stepper motor for dyn. cornering light switched off

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Diagnostic conditions

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Fault setting condition(s)

Incorrect coding (pin assignment/contacts) of the headlight control unit connector could be the cause.

- **Control module** Headlights **Pin assignment** incorrect
- **Control module** Headlights **Pin assignment** interchanged

Fault effects

- ◆ System switches off
- ◆ Display in instrument cluster

Diagnostic information - Headlights

Control module Programming

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Fault setting condition(s)

- **Control module** Headlights incorrectly programmed
- **Control module** Headlights not programmed
- Incorrect software version left right
- incorrect hardware version left right
- **Control module** not in operating mode

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Diagnostic information - Headlights

Adaptive Frontlighting System

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NOTE

Among other things, the **Adaptive Frontlighting System** (AFS) is a dynamic light distribution and is produced by a cylinder that can rotate. It possesses three different positions/contours to change the light/dark limit for ideal highway, motorway and poor-weather lighting.

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- Leads Contacts
- **Servo motor Adaptive Frontlighting System (Headlight housing)**

Fault effects

- ♦ **Adaptive Frontlighting System** Function not available or available only to a limited extent

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Coding plug Wrong connector assignment

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Diagnostic conditions

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Fault setting condition(s)

Incorrect coding (pin assignment/contacts) of the headlight control unit connector could be the cause.

- **Control module** Headlights **Pin assignment** incorrect
- **Control module** Headlights **Pin assignment** interchanged

Fault effects

- ◆ System switches off
- ◆ Display in instrument cluster

Gas discharge lamp, xenon, power supply

WARNING

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- ◆ Wait an appropriate amount of time for cooling
- ◆ Wear suitable protective gloves to protect your hands.

CAUTION

Danger of injury from high voltage!

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Diagnostic conditions

- Headlight (xenon) control unit internal function test
- Ignition on

Possible fault causes

- ◆ Wiring harness in headlight faulty (headlight housing)
- ◆ Headlight control unit faulty

Gas discharge lamp (xenon)

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Diagnostic conditions

- **Headlight** (xenon) control unit internal function test
- Ignition on

Possible fault causes

- ◆ Line connection from headlight control unit to gas discharge lamp (xenon) faulty
- ◆ Gas discharge lamp (Xenon) faulty
- ◆ Headlight control unit faulty

Gas discharge lamp (xenon)

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- ◆ Line connection from headlight control unit to gas discharge lamp (xenon) faulty
- ◆ Gas discharge lamp (Xenon) faulty
- ◆ Headlight control unit faulty

Diagnosis information - (Xenon) gas discharge igniter

Control module Power supply

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NOTE

NOTE

Sporadic faults!

inactive Fault - Erase fault memory and read it out

Diagnostic conditions

- Headlights switched on
- Ignition on

Fault setting condition(s)

- Lines/connections closed Light bulb Faulty
- Signal Terminal 15
- Signal Terminal 56b

Fault effects

- ♦ System switches off, Headlights **OFF**
- ♦ Display in instrument cluster

Diagnostic information - Headlights

Coding plug Wrong connector assignment

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- **Control module** Headlights **Pin assignment** interchanged

Fault effects

- ◆ System switches off
- ◆ Display in instrument cluster

Diagnosis information - (Xenon) gas discharge igniter

Control module Power supply

WARNING

HBA Xenon: Electric shock and burns from high voltage.

- ◆ The lights should always be switched off before removing and installing the headlights.
- ◆ Only operate the gas-discharge lamp when installed in the headlight.
- ◆ Carry out repair work on the headlight or wiring only with the component removed and separated from the voltage supply.

NOTE

NOTE

Sporadic faults!

inactive Fault - Erase fault memory and read it out

Diagnostic conditions

- Headlights switched on
- Ignition on

Fault setting condition(s)

- Lines/connections closed Light bulb Faulty
- Signal Terminal 15
- Signal Terminal 56b

Fault effects

- ◆ System switches off, Headlights **OFF**
- ◆ Display in instrument cluster

Diagnostic information - Control module Headlights

Vehicle identification, different Vehicle Identification Number (VIN)
Vehicle Identification Number (VIN) , implausible

CAUTION

Danger of injury from high voltage!

- ◆ The light must always be switched off when removing and installing the headlights.
- ◆ The gas-discharge lamp may only be operated when installed in the headlight.
- ◆ Only carry out repairs to the headlight or wiring when the part is removed and disconnected from the power supply.

INFORMATION

- ◆ Calibration is not possible with different/implausible **Vehicle Identification Number (VIN)** .

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- Internal function test on headlight (Xenon) control unit

NOTE

This fault is set when the Vehicle Identification Number (VIN) entered in the control unit is not the same as the one reported via CAN.

A new control unit receives the Vehicle Identification Number (VIN) when it is connected for the first time.

Fault setting condition(s)

- Coded Vehicle Identification Number (VIN) in the control unit is not the same as the vehicle's VIN sent via CAN.
 - Control unit is new, used for the first time or vehicle handover not yet performed.
 - Control unit comes from another vehicle.
 - Vehicle Identification Number (VIN) not entered correctly in the **Control module** Headlights .
- Check and correct the Vehicle Identification Number (VIN) of the **Control module** Headlights (in the PIWIS Tester ⇒ **Control module** Headlights ⇒ Extended identification ⇒ Chassis number)

Fault effects

- ◆ System does not function
- ◆ Display in instrument cluster

HBA stepper motor



CAUTION

Danger of injury from high voltage!

- ◆ The light must always be switched off when removing and installing the headlights.
- ◆ The gas discharge lamp may only be operated when installed in the headlight.
- ◆ Only carry out repairs to the headlight or the wiring when the part is removed and disconnected from the power supply.

Diagnostic conditions

- Headlight (xenon) control unit internal function test
- Ignition on

Possible fault causes

- ◆ Line connection or contacts to HBA servo motor faulty
- ◆ HBA stepper motor overheated
- ◆ HBA stepper motor mechanically blocked
- ◆ HBA stepper motor faulty
- ◆ **Headlight** control unit coded incorrectly

Diagnostic information - Headlight control unit

Control unit, coded incorrectly

CAUTION

Danger of injury from high voltage!

- ◆ The light must always be switched off when removing and installing the headlights.
- ◆ The gas-discharge lamp may only be operated when installed in the headlight.
- ◆ Only carry out repairs to the headlight or wiring when the part is removed and disconnected from the power supply.

Diagnostic conditions

- Headlight control unit coded incorrectly
- Ignition on

Possible fault causes

- ◆ Line connection from **headlight** control unit to headlight housing connector faulty
- ◆ Wiring at headlight control unit connector Pin 5 and Pin 6 interchanged
- ◆ Headlight control unit faulty